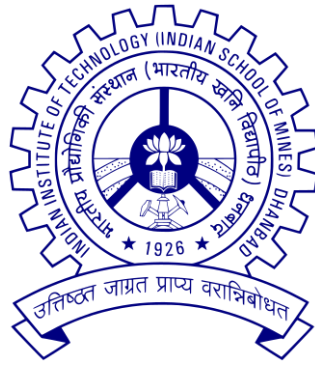




SYSTEM-LEVEL PROPOSAL: NEXT-GEN FEATURES, FAST CHARGING & TPMS INTEGRATION IN A COMMERCIAL THREE-WHEELER ELECTRIC VEHICLE

Project: Bajaj Campus Challenge 2026 – Let’s rEVolve

Vehicle Category: Commercial Three-Wheeler Electric Vehicle (L5 Category)

TEAM PROFILE :

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Undergraduate Electronics and Communication Engineering student at IIT ISM Dhanbad with experience in embedded systems, system design, and C++. Actively working on electric vehicle technologies and real-time monitoring systems to improve vehicle safety, efficiency, and operational performance.

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Undergraduate student at IIT ISM Dhanbad with academic interests in Power Electronics and Machine Learning. Focused on applying intelligent, data-driven approaches to energy-efficient and sustainable electric vehicle systems.

Executive Summary

This proposal presents a cost-effective system-level integration of next-generation vehicle features, fast charging, and a Tire Pressure Monitoring System (TPMS) for a commercial three-wheeler electric vehicle. The design emphasizes safety, uptime, scalability, and low system complexity to meet the operational demands of commercial mobility.

Identification and Justification of Next-Generation Features

A lightweight predictive maintenance framework replaces heavy digital-twin models by using independent ML modules to detect early failures in critical EV components. The goal is repair assistance, preventive replacement, and improved uptime.

Battery Thermal Runaway Prediction

- Monitors temperature rise rate, voltage sag, and internal resistance trends
- Flags abnormal heat buildup and cell degradation
- Enables early battery swap before catastrophic failure

High-Voltage Contactor Health Monitoring

- Tracks contact resistance, switching delay, and coil current signature
- Detects mechanical and electromagnetic wear
- Prevents unexpected breakdown during DC charging

DC-DC Converter Aging Detection

- Analyzes output ripple and efficiency drift
- Identifies capacitor/inductor degradation
- Supports preventive replacement scheduling

This modular approach improves safety, reliability, and maintenance planning without adding system complexity, making it suitable for cost-sensitive commercial EV deployment.

Fast-Charging Standards Evaluation

Since CCS2 introduces a voltage mismatch and Type-7 hybrid charging would require major architectural restructuring, Type-6 DC charging is selected as the most feasible and cost-effective solution. The design adds a parallel DC fast-charging pathway while preserving the existing AC overnight charging system.

Core Components :

DC Contactor

- Acts as a safety gate between charger and battery
- Closes only after successful CAN handshaking
- Supports controlled pre-charge and main contact sequencing

Electronic Vehicle Communication Controller (EVCC)

- Handles CAN protocol translation between BMS and charging point (CHAdeMO/Type-6)
- Executes pre-charge and main contactor firing logic
- Activates dedicated thermal management during fast charging

Dedicated Thermal System

- Aluminum battery enclosure with integrated cooling fins
- Lightweight wind-tunnel casing for directed airflow
- High static-pressure fans enable forced convection cooling
- Maintains safe battery temperature under rapid charging loads

This parallel architecture enables safe high-power charging while maintaining compatibility with existing infrastructure and minimizing system redesign.

TPMS Architecture & Conceptual Design

The TPMS is a critical component for maintaining range efficiency and safety in EVs.

System Architecture

- **Sensor Layer:** Three wireless internal sensors (1 Front, 2 Rear) that measure pressure and temperature every second during motion.
- **Communication:** BLE (Bluetooth Low Energy) is utilized for data transmission to the Vehicle Control Unit (VCU) to reduce wiring and harness complexity.
- **Real-time Display:** Current PSI/Bar values for all three tires are displayed on the main dashboard, with a "Low Pressure" visual alarm.

Tire-Position Identification

- **Methodology:** The system uses **Signal Strength (RSSI) localization**.
- **Logic:** The receiver is mounted near the rear axle; the system identifies the front tire by its lower signal strength and distinguishes the left/right rear tires based on pre-set sensor IDs during the manufacturing EOL (End of Line) calibration.

Design Assumptions and Technical Parameters

The proposed system is developed based on practical operating assumptions representative of commercial three-wheeler EV usage in Indian urban environments:

- **Battery Capacity:** 9 kWh lithium-ion battery pack optimized for a 100–120 km operating range
- **Charging Capability:** 3.3 kW AC slow charging with support for up to 10 kW Type-6 DC fast charging
- **Duty Cycle:** High-frequency start–stop operation typical of commercial service
- **Usage Pattern:** Urban commercial L5 vehicle with daily high-utilization cycles

6.Cost-Benefit & Feasibility Analysis

Category	Component Upgrade	Estimated Add-on Cost (per unit)	Expected Benefit / ROI
Safety & Life	Predictive ML (Thermal/Contactor)	₹1,500 - ₹3,000	25% reduction in unplanned downtime; 15% longer battery life.
Charging	Type-6 DC Architecture + EVCC	₹12,000 - ₹18,000	Charging time reduced from 6 hrs to <1 hr; 2x daily trip capacity.
Thermal	Aluminum Casing & Wind Tunnel	₹5,000 - ₹8,500	Prevents power de-rating in 40°C+ heat; enables safe DC fast charging.
Safety	Wireless TPMS (RSSI Logic)	₹3,500 - ₹5,500	5-8% improvement in range efficiency; 30% reduction in tire wear.